

Referee Report for Economic Journal Manuscript 20251300, “Trading Social Status for Genetics in Marriage Markets: Evidence From Great Britain and Norway”

I previously refereed this paper for the Journal of Political Economy. I have read the revised version, and most of my comments still apply. Thus I have retained much of the language from my original report, although I have expanded some points and added some new comments.

This submission develops and tests a model of assortative mating in which spouses match based on a linear index of both socio-economic status (SES) and genetic factors (G). The central theoretical result is that, across successive generations, this kind of assortative mating will produce a positive correlation between higher SES and the genetic factors that are valued in the marriage market. At an individual level, this means that shocks to socio-economic status can causally affect the average genetic endowments of one’s descendants. At the population level, these dynamics mean that the distribution of genetic endowments may be affected by institutions that shape the joint distribution of SES and G. The authors stress that the model provides a new account for the emergence of genetic correlations with SES. The literature on social science genetics has long dealt with gene-SES correlations - but this is often interpreted as a result of meritocracy - the result of genes causally affecting SES within a generation, with the next generation inheriting both higher SES from their parents, and the advantageous genetic endowments that may be (partially) responsible for that higher status. By contrast, the mechanism described in this paper generates a gene-SES correlation from the causal effect of higher SES on the genetic endowments of one’s spouse, both of which are passed along to the next generation.

The empirical portion of this paper tests the mechanism outlined in the paper by examining the effect of an exogenous shock to socioeconomic status: birth order. Using samples from both the UK and Norway, the authors exploit birth order as an exogenous shock to own SES. The authors replicate existing results on the effects of birth order on own education,

income, IQ, height, BMI, and health. Then the authors show that birth order is negatively associated with the PSEA (polygenic score for educational attainment) of an individual's spouse. That is, individuals who are born earlier become matched to spouses with higher values of a genetic index that predicts educational attainment. The authors interpret this as evidence that higher SES causally affects the genetic endowments of one's spouse (and thus children). Mediation exercises suggest that own education plays an important role as a mechanism for this result in both countries, with income emerging as an additional important mediator in the UK but not Norway.

I read this submission with great interest, and I believe that the results here have the potential to be quite valuable to the literatures on genetic endowments, and the broader literature on intergenerational persistence. The model provides an interpretation of gene-SES correlations that has been under-appreciated or ignored by the broader literature (at least in my reading). The empirical analyses are interesting, and I very much appreciate the effort in replicating the results in multiple contexts. However, I have a few concerns about the analyses conducted here. Broadly, these can be classified as 1) concerns about the overall contribution of the findings here for an economics audience, and 2) concerns about the execution of the empirical analyses and their links to the theoretical claims.

Comments:

- **Connection between Theory and Empirics**

The authors stress that the model contains a new mechanism - in the marriage market people make trade-offs between genes and social status when assessing the attractiveness of potential spouses. Much of the motivation and discussion of the model suggests that x_1 (genetics) and x_2 (SES) reflect distinct traits that are separately valued in a spouse. This creates the potential for a true trade-off between distinct attributes when selecting a partner. Indeed, the references to Austen and Marx evoke a trade-off between two or more distinct traits (e.g. money and physical beauty) that people very clearly independently value. The empirical analysis in this paper does not offer a sat-

isfying test of this theory because the measure of spouse’s genetics studied here (the polygenic score for Educational Attainment) is so closely tied to the spouse’s SES. The empirical results seem perfectly consistent with a standard assortative mating model in which spouses match on one trait - human capital - which is correlated with education, income, and the genetic factors measured by the polygenic score for education. If individuals are only observing and matching on one dimension of attractiveness (human capital), it would still be the case that the positive shock from birth order would boost an individual’s own human capital and thus increase the human capital of their spouse. Since the polygenic score for education is positively correlated with human capital, we would expect that having low birth order would thus increase the expected polygenic score (for education) of an individual’s spouse. Given that the empirical results are quite consistent with a standard assortative mating model, it is not clear that they offer evidence in favor of the SGAM mechanism proposed here.

One constructive suggestion I would have for the authors is to expand their empirical analysis and look at a broader set of spousal polygenic scores that include some that are less tightly tied to human capital / SES, and which more plausibly capture traits that are distinctly valued in the marriage market. For example, the authors could look at polygenic scores related to personality (e.g. extraversion), physical activity, BMI, and others. Presumably, given the predictions of the model, one should see a positive effect of low birth order on the values of these genetic indices as well.

- **Engagement with Existing Literature:** Previously, I suggested that the authors cite Rustichini et al 2023 (Educational Attainment and Intergenerational Mobility: A Polygenic Score Analysis), published in the JPE. The authors do cite this paper in passing, but I believe they need to provide a more thorough discussion of that paper, how it is related to their own work, and the contribution of the analyses in this manuscript given what is in that paper. The present manuscript is written in a way that suggests that SGAM provides a completely new framework for thinking about the

correlation between SES and genetics. While there are certainly novel results in the manuscript, the modelling framework developed here shares many similarities with the one developed by Rustichini et al (2023). For example, in that paper, the attractiveness of an individual is explicitly determined as a linear function of a genetic factor and income. The parameter w_y governs the relative preference given to income v.s. genes, much the same way that the a parameter functions in the current manuscript. So the model developed by Rustichini et al (2023) features an assortative mating structure that is quite similar to the one being proposed here, where genes and a socioeconomic outcome both contribute to attractiveness in the marriage market and can be “traded-off” against one another in the marriage market. When discussing the contribution of the manuscript, the authors should be precise about the nature of the theoretical contribution. How much of the contribution here is a fundamentally new modelling framework for genes and assortative mating, and how much comes from implications of modelling frameworks already present in the literature?

- **Framing of the Trade-off between Genes and SES**

I find it unhelpful for the authors to frame the central trade-off in the model as one between “genes” and “SES.” Nearly all complex traits that humans care about have some genetic influences. This framing suggests that SES itself does not have a genetic component, when in fact the huge amount of work done in the social science genetics literature has focused on educational attainment and other outcomes that are clearly linked to SES. Rather, it seems like a better starting place is to think about a model of assortative mating where individuals value multiple traits - all of which are influenced by genetic factors. Such a framework would allow for more precise arguments about genetic correlations - an SES shock arising from birth order, for example, might positively affect the expected value of the genetic endowments in one’s spouse at different rates depending on whether you are talking about genes related to SES, education, physically attractive traits, personality characteristics, etc.

- **Clarifying the Contribution:** Reading the paper, I am left with the impression that there is potential here for a very important contribution, but I feel that it is never quite clearly stated or demonstrated. The authors develop an interesting contrast between gene-SES correlation arising from assortative mating (as in their model), compared to a scenario in which it arises because of effects on SES and simultaneous transmission of SES and genes (the authors refer to this as meritocracy). This could be very interesting if the authors could provide a reason why a general economics audience should care about which of these scenarios accounts for gene-SES correlation. For example, are models of intergenerational mobility (or the effects of policy on such mobility) producing misleading conclusions by emphasizing meritocracy as the source of gene-SES correlation? I do not feel that the current manuscript clearly lays out how the SGAM mechanism resolves controversies or significantly changes the conclusions that economists tend to draw about inequality or policy addressing intergenerational mobility. For example, on p. 5, in explaining the relevance of their findings for the literature on mobility, the authors cite a debate about whether or not empirical estimates of parent-offspring correlations are consistent with longer-run, multi-generational correlations, or correlations with more distant relatives. Yet it is not clear to me whether the results presented in the paper help resolve this or shed more light on reconciling the literature. Similarly, the authors mention a “conventional wisdom” that assortative mating on SES increases cross-sectional inequality. But it is not clear at all that the results presented here overturn this conventional wisdom. Overall, I think the authors need to make a clearer case for why these results matter to a more general economic audience.
- **Questions about the Empirical Analysis:** I have several comments and concerns related to the ways in which the empirical results are implemented and interpreted.

1. **Birth Order and Parental Age:** It is noteworthy that in the UK sample, a significant negative association between birth order and spousal PSEA emerges

only when controlling for parental age. A similar pattern is present in the Norwegian sample - although there is a negative relationship between birth order and spousal PSEA even without parental age controls, it becomes dramatically larger after controlling for parental age. This is not necessarily problematic, but it could signal that the results (at least their magnitudes in the case of the Norwegian data) could be an artifact of the econometric specification. For example, the negative coefficient on birth order could arise if the true data generating process features non-linearities in parental age. Then in a mis-specified version of the model that restricts the relationship to be linear, the model may be fitting a non-linear relationship by assigning a (more) negative coefficient to another variable that is positively correlated with parental age. I would recommend that the authors present specifications which flexibly allow for non-linearities in the relationship between parental age and spousal PSEA. It is also not clear why parental ages are being averaged. Father's and mother's age could clearly differ in their impact on SES (certainly there are different biological considerations), and given the apparent importance of controlling for parental age in the empirical results, it may be important to more thoroughly explore these specification choices.

2. **Consistent Sample Sizes:** Related to the previous comment, it is also noteworthy that in Table 2, the sample size drops significantly when controlling for parental age, presumably because of missing values. Separate from the issue of the correlation between parental age and birth order, it would be helpful to report Table 2 on a consistent sample to understand how much of the instability of the coefficient estimate on birth order is a consequence of the changing sample and how much is a result of a correlation between parental age and birth order. I know this is not an issue for the Norwegian sample.
3. **Birth Order - Effects on Personality:** An existing literature reports effects of birth order on personality, including emotional stability and social ability (Black

et al 2018, ReStat “Born to Lead?”). The existence of such effects would cloud the interpretation of the empirical analyses as tests of the theory. If birth order affects not only SES, but also personality traits that could affect attractiveness in the marriage market, then it is difficult to interpret the negative effects of birth order on spousal PSEA as demonstrating the effect of SES shocks on spousal genes rather than the effects of other personality traits.

4. **Timing of Mediator Measures:** One issue with interpreting the mediation exercise is that some of the mediators are likely to be measured after marriage, and thus could reflect causal effects of the marriage (and spousal characteristics). For example, in the UK sample, I am assuming that BMI, health, and fluid IQ are measured later in life after the marriage. This raises the question of whether associations between these mediators and spousal PSEA reflect returns to these mediating characteristics in the marriage market, or the effects of higher or lower spousal PSEA on own outcomes.